# Three complex initial envelopes

A bounded spatial Dysthe simulator demonstration

16 October 2026

## 1 Problem and prescribed inputs

The simulator advances the scaled complex free-surface envelope $u(\xi, \tau)$, where $\xi$ is propagation distance and $\tau$ retarded time:

$$u_\xi = -\mathrm{i}u_{\tau\tau} - \mathrm{i}|u|^2 u - \epsilon\big(8|u|^2 u_\tau + 2u^2\overline{u_\tau} + 2\mathrm{i}u|D_\tau|(|u|^2)\big).$$

The model contract is `water-wave-dysthe-spatial-v1`. The coefficients are $(1, 1, 8, 2, 2)$. The periodic interval is $[-80, 80)$, $N = 4096$, $\epsilon = 0.025$, $\Delta\xi = 0.00125$, and $0 \le \xi \le 4$. There are 81 saved positions. The Fourier method retains $|m| < N/4$, uses mean-flow symbol $|k|$ with zero mean mode, and applies Lawson integrating-factor RK4.

Let $F$ be the recomputed periodic branch with $\mu = 2$, $s = 0$, and full traveling residual $2.03\times10^{-11}$. The three initial fields are $F(\tau)$; $F(\tau)[1+0.02\exp(-((\tau-1.3)/2)^2)]$; and $\exp(-\tau^2/8)\exp(0.3\mathrm{i}\tau)$. The second is a shifted localized amplitude perturbation. The third has amplitude 1, width parameter 2, wavenumber 0.3, and zero chirp. The plots retain physical coordinates and show envelope amplitude, rather than reconstructed surface elevation.

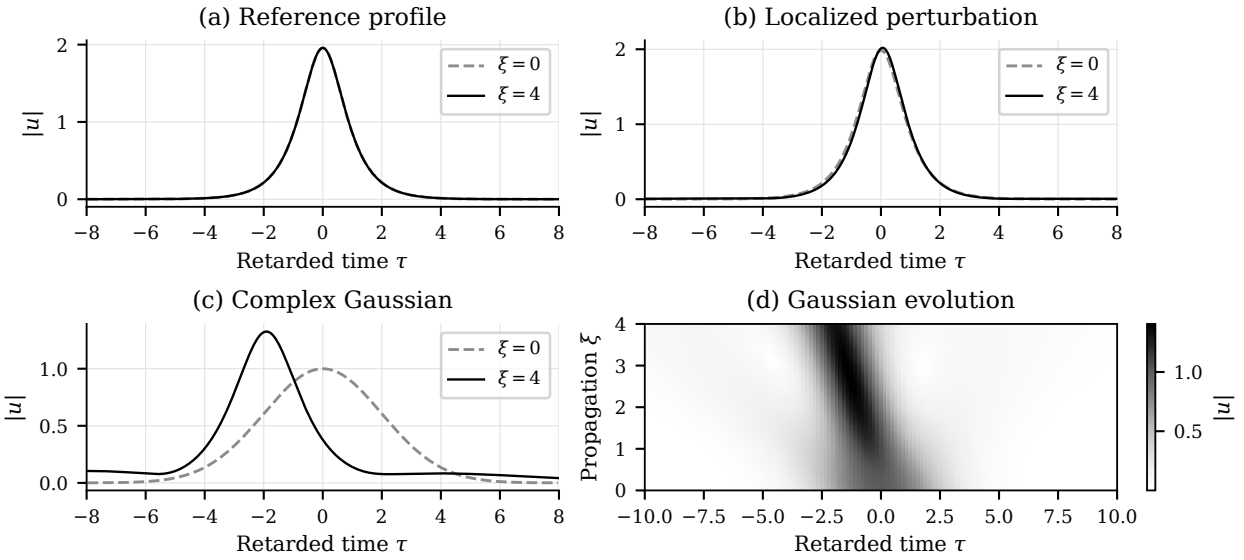



Figure 1: Initial and final amplitudes for all three cases, and the Gaussian trajectory. These are central-window views of the saved full-domain complex fields.

## 2 Observed evolution

The reference keeps its amplitude and width to the reported accuracy while its fitted global phase changes by $-8.000000$ radians. Its final shape distance is $2.77 \times 10^{-10}$. The perturbed profile deforms modestly and develops exterior action. The Gaussian focuses to maximum amplitude 1.406646 and ends at amplitude 1.325587. Its final shape distance is 39.31% despite a net width change of only 0.0197%. Width alone therefore misses substantial complex-envelope evolution.

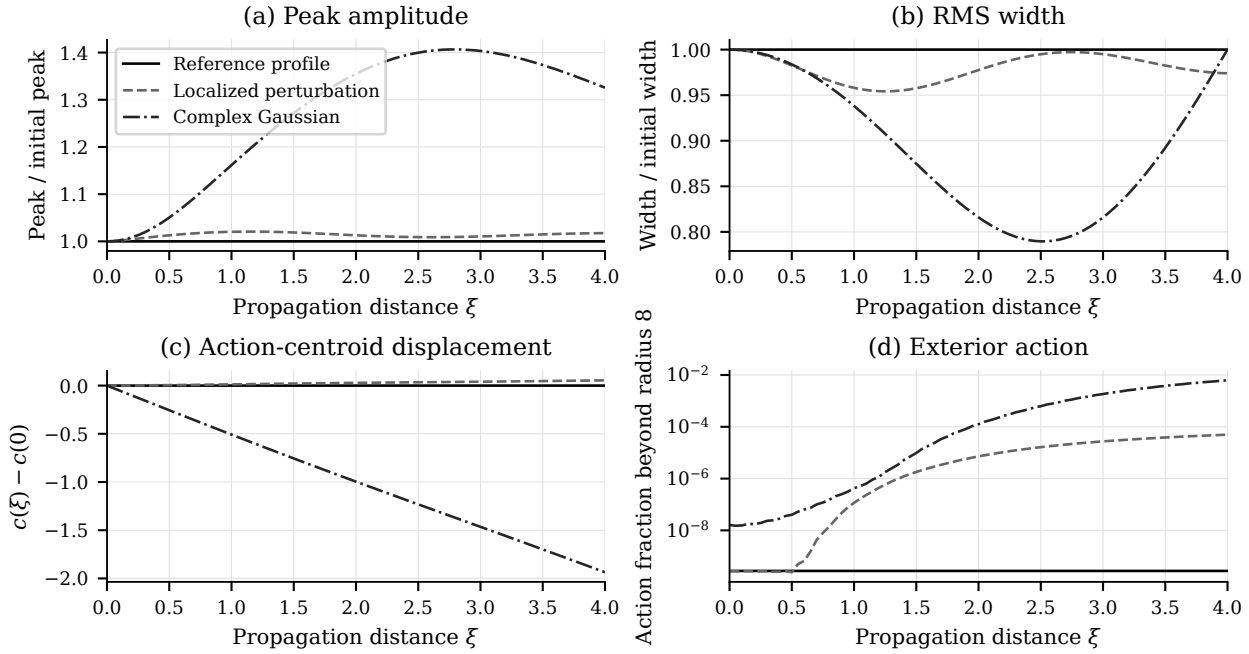



Figure 2: Peak and width relative to their own initial values; action-centroid displacement; and action beyond periodic distance 8 from that centroid. The exterior fraction includes packet tails and spreading and is not automatically radiation.

Table 1: Final values at $\xi = 4$. Shape is the relative complex L2 distance to each case's initial field after removing translation and constant phase, without amplitude rescaling.

| Case | Peak | RMS width | Displacement | Shape | Tail fraction |
|---|---|---|---|---|---|
| Reference profile | 1.958116 | 0.623359 | $-3.21 \times 10^{-10}$ | $2.77 \times 10^{-8}$% | $2.72 \times 10^{-10}$ |
| Localized perturbation | 2.018419 | 0.606491 | 0.054528 | 2.57% | $4.96 \times 10^{-5}$ |
| Complex Gaussian | 1.325587 | 1.414492 | -1.936173 | 39.31% | $6.22 \times 10^{-3}$ |

The centroid is obtained from the first circular action moment; displacement is unwrapped relative to its initial value. RMS width is the action-weighted shortest periodic distance about that centroid. These statistics become ambiguous for broad or wrapped fields. Here circular concentration remains above 0.998. The complex shape diagnostic also detects phase variation that amplitude plots cannot show.

## 3 Numerical qualification and reproducibility

Each default case has independent comparisons with $N = 8192$ at fixed domain, $\Delta\xi = 0.000625$ at fixed grid, and $(L, N) = (320, 8192)$ at fixed spacing. All nine comparisons and the recorded design screens pass. Profiles are recomputed for each period. The table reports maxima over all 81 saved positions, retaining complex phase in the raw comparisons.

Table 2: Maximum relative complex L2 differences. Domain comparisons use common physical coordinates in the central $[-80, 80)$ window. The last column additionally removes translation and constant phase.

| Case | Grid | Step | Domain | Domain orbital |
|---|---|---|---|---|
| Reference profile | $4.17 \times 10^{-15}$ | $3.07 \times 10^{-9}$ | $1.54 \times 10^{-6}$ | $1.54 \times 10^{-6}$ |
| Localized perturbation | $3.09 \times 10^{-15}$ | $4.08 \times 10^{-9}$ | $2.26 \times 10^{-6}$ | $1.83 \times 10^{-6}$ |
| Complex Gaussian | $9.28 \times 10^{-16}$ | $4.05 \times 10^{-11}$ | $2.18 \times 10^{-5}$ | $9.66 \times 10^{-9}$ |

The raw-field normalization is the refined field norm at each position. The chosen gates are $2\times10^{-4}$ for raw differences and $10^{-4}$ for orbital differences and signed-shape discrepancies. The latter compare each solution's deformation from its own initial field. Their additional relative response gate is 0.02, used only above a $10^{-7}$ field-norm response floor. The tiny reference response falls below that floor. Two levels demonstrate agreement; they provide neither an observed convergence order nor a rigorous error bound. Periodic mean flow can depend on $L$ even when boundary action is tiny; domain differences include that dependence.

Signed-shape comparisons can have an approximately $10^{-8}$ fitting floor under the inherited orbital optimizer even when raw grid differences are near $10^{-15}$. This is a diagnostic precision limit, not evidence of an unresolved propagated field. Action conservation is supporting evidence; padded continuum energy and momentum are not assumed exact invariants of the projected equation or RK map.

The stored arrays `tau`, `xi`, `u`, and `initial` are available in each case's `fields.npz`; `u` is complex128 of shape $(81, 4096)$. Reports record parameters, definitions, qualification, source hashes, and field checksums. Plotting verifies those checksums and metadata against the saved summary before reading the fields. The completed 12-trajectory campaign, including refinements, used 106.3 seconds of solver/diagnostic time and peak process RSS 197.0 MiB.

From the task root, reproduce these plots and tables without rerunning propagation:

```
python3 stage2/plot_demo.py
cd stage2/report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The plot manifest, source data, and earlier numerical-reference report are:

```
stage2/results/plot-manifest.json
stage2/results/demo/
stage2/review/dysthe-stage-one-reference.pdf
```

These checks qualify only the exact three defaults and $\xi \in [0, 4]$. Custom parameters and longer horizons require their own refinements. This small simulator demonstrates varied complex initial envelopes and preserves their full fields; it establishes no soliton discovery, stability theorem, Euler-model validation, training dataset, or learned operator.